

Momentum Variations Research IV

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Abstract

This memo investigates two extensions to the standard momentum strategy currently implemented by Silver Fund: idiosyncratic momentum and volatility-scaled momentum. We evaluate idiosyncratic momentum signals constructed from four factor models—CAPM, Fama-French 3-Factor, Fama-French 5-Factor, and the Barra model—and find that signal performance improves monotonically with the explanatory power of the underlying factor model, with Barra idiosyncratic momentum achieving the highest Sharpe ratio among residual-based signals. We then apply volatility scaling to the top-performing signals and show that it improves risk-adjusted returns for both standard and idiosyncratic momentum. While standard volatility-scaled momentum achieves the highest full-sample Sharpe ratio, we demonstrate that volatility-scaled Barra idiosyncratic momentum exhibits substantially stronger performance during the COVID-19 crash period, consistent with its reduced exposure to systematic factor reversals. Based on these findings, we recommend replacing Silver Fund's current standard momentum signal with volatility-scaled Barra idiosyncratic momentum.

Sources

Blitz, D., Pang, H., & van Vliet, P. (2013). The idiosyncratic momentum anomaly. *Journal of Business & Economic Statistics*, 31(1), 44–56. <https://doi.org/10.1080/07350015.2012.723155>

Hanauer, M. X., & Lauterbach, R. (2023). Enhanced momentum strategies. *Journal of Banking & Finance*, 154, 106939. <https://doi.org/10.1016/j.jbankfin.2023.106939>

Previous Work

Last semester, Andrew worked on replicating both “The idiosyncratic momentum anomaly” and “Enhanced momentum strategies.” Results showed that idiosyncratic momentum and volatility-scaled momentum both outperformed standard momentum in and out of sample. In this memo, we investigate two questions:

1. Which is the best factor model for computing residual returns?
2. What is the compound effect of volatility scaling in addition to idiosyncratic momentum?

Economic Intuition

The economic intuition for the momentum anomaly has been extensively discussed within Silver Fund, specifically the underreaction hypothesis. In this memo, we focus on the rationale for idiosyncratic and volatility-scaled momentum. Standard momentum, as currently implemented by Silver Fund, uses total return momentum, implying that higher-momentum stocks may be high-momentum because the underlying factors they're exposed to outperform. In reality, the underreaction hypothesis primarily concerns investors' underreaction to firm-specific information. By using the idiosyncratic component, we can isolate this effect without incorporating unwanted factor risk. Furthermore, volatility scaling addresses the crash risk of standard momentum while limiting the strategy to selecting stocks solely on the basis of high volatility.

Factor Models

In calculating the idiosyncratic component, we consider the four different factor models familiar to Silver Fund. These factor models are given below:

Capital Asset Pricing Model:

$$R_{it} - R_{ft} = \alpha_i + \beta_i(R_{mt} - R_{ft}) + \epsilon_{it}$$

Fama-French 3-Factor Model:

$$R_{it} - R_{ft} = \alpha_i + \beta_{i1}(R_{mt} - R_{ft}) + \beta_{i2}SMB_t + \beta_{i3}HML_t + \epsilon_{it}$$

Fama-French 5 Factor Model:

$$R_{it} - R_{ft} = \alpha_i + \beta_{i1}(R_{mt} - R_{ft}) + \beta_{i2}SMB_t + \beta_{i3}HML_t + \beta_{i4}RMW_t + \beta_{i5}CMA_t + \epsilon_{it}$$

Barra Factor Model:

$$R_i = \sum_{j=1}^K \beta_{ij} f_j + \epsilon_i$$

In our research, we examined the performance of idiosyncratic momentum across these factor models. We recognized that using a residual from a factor model with fewer factors would inherently trade momentum on the excluded factors. However, we didn't know if that would ever be desired or if we should always use Barra. When consulting with Brandon, we received the following explanation for why the idiosyncratic component from Barra is superior:

Suppose the true model of returns is $r_{it} = \sum_{j=1}^k \beta_{ij} f_{jt} + e_{it}$. So, you have $k + 1$ things you could forecast if you knew the true model. Suppose there are Q possible actual factor models to choose from to estimate and you choose model q . Suppose your proposed factor model is $r_{it} = \sum_{j=1}^{m^q} \gamma_{ij}^q g_{jt}^q + u_{it}^q$, giving you a maximum of $m^q + 1$ things to actually forecast. You can create a momentum signal on those $m^q + 1$ individually or any version of $\sum_{j \in J} \gamma_{ij}^q g_{jt}^q + u_{it}^q$, where J is the set of all subsets of natural numbers up to m^q and including or excluding the "idiosyncratic" term, u^q . The cleanest version is computing [signal] separately on each factor and on the idiosyncratic term. Computing [signal] on any of the subsets assumes that the loadings g^q are the same as the optimal signal weights you would have used to combine the signals computed on each component separately, which is weakly suboptimal.

The next point is which $q \in Q$ to pick. Pick the one that best explains the contemporaneous cross-section of returns, which is Barra.

Signal Construction

1. Momentum

The standard momentum signal is constructed by taking the daily log returns and applying a 230-day rolling sum for each stock. This value is then shifted by 21 days to create the desired $t_{12} - t_2$ time period. See Figure 1.

The following step applies to the ensuing idiosyncratic (excluding Barra) and volatility-scaled signal constructions: In order to extract the idiosyncratic portion of each signal, perform a rolling OLS regression using the desired factor model (CAPM, FF3, FF5) to create each factor's beta coefficients to be used in the next steps.

2. Idiosyncratic Momentum (CAPM, FF3, FF)

Using the beta coefficients from the rolling regression, take each stock's excess return and subtract off the return that is predicted by each specific model's factors. Then, perform the standard momentum signal on these values. See Figure 2.

3. Idiosyncratic Momentum (Barra)

Using the Barra data, take the daily log of the specific return for each stock and perform a rolling sum over 230 days. Subsequently, shift that value by 21 days. See Figure 3.

4. Volatility Scaled Momentum (Momentum)

This signal follows the same process as all other signal constructions, but adds the step of dividing each rolling return by its rolling 230-day standard deviation before shifting by 21 days. See Figure 4.

Experiments

We constrain the portfolios to have a zero beta relative to the market and to have portfolio weights that sum to zero. A \$5 price filter is applied prior to portfolio construction. For each signal, we pick a gamma that achieves approximately 5% active risk over the period from 1996-07-31 to 2024-12-31. After computing the weights for every trading day in the full sample, we use the weights and forward returns for the full sample period to highlight the performance of each signal. To not include an abundance of results, we show the full sample backtest of the non-volatility-scaled signals, the full sample backtest of the volatility-scaled signals for the top 2 signals from the non-volatility-scaled set, and the COVID-19 sample backtest for the volatility-scaled signals.

1. Non-volatility-scaled

We first illustrate the performance of each signal without volatility scaling. We find that the standard momentum signal has the highest Sharpe ratio. Importantly, we find that, among idiosyncratic momentum signals, increasing the number of factors in the model increases the strategy's Sharpe ratio. This supports our earlier theory, showing that the best factor model for idiosyncratic momentum is the Barra model, as it has the highest explanatory power for cross-sectional returns. See Figures 5 and 6.

2. Volatility-scaled (Top 2)

We now present the performance of the standard momentum and Barra idiosyncratic momentum signals after applying volatility scaling during signal construction. We find that volatility scaling improves the performance of both signals, but the standard momentum signal still has the highest Sharpe ratio. In our next test, we will illustrate why the Barra idiosyncratic momentum signal will be preferable to the standard momentum signal despite having a lower Sharpe ratio. See Figures 7 and 8.

3. COVID-19 Sample Period

One reason idiosyncratic momentum is preferable to standard momentum is that it is less affected by market crashes, such as the COVID-19 pandemic. We present this for the sample period 2019-01-01 to 2022-12-31, which fully encompasses the months before and after the COVID-19 crash. We find that the idiosyncratic momentum strategy achieves a much higher Sharpe than the standard momentum strategy. See Figures 9 and 10.

Conclusion

In conclusion, we find that signal performance improves monotonically with the explanatory power of the underlying factor model, volatility scaling improves all momentum variations, and idiosyncratic momentum outperforms standard momentum during the COVID-19 pandemic. Accordingly, we vote to replace the current standard momentum signal with volatility-scaled Barra idiosyncratic momentum in the Silver Fund portfolio.

Appendix

1. Standard Momentum Signal

```
momentum.py

def momentum() → pl.Expr:
    return (
        pl.col("return")
        .log1p()
        .rolling_sum(230)
        .shift(21)
        .over("barrid")
        .alias("momentum")
    )
```

2. Fama-French 3 Idiosyncratic Momentum Signal

```
ff3_idiosyncratic_momentu

def ff3_idiosyncratic_momentum() → pl.Expr:
    return (
        pl.col("return_rf")
        .sub(
            pl.sum_horizontal(
                pl.col("B_const"),
                pl.col("B_mkt_rf").mul("mkt_rf"),
                pl.col("B_smb").mul("smb"),
                pl.col("B_hml").mul("hml")
            )
        )
        .log1p()
        .rolling_sum(230)
        .shift(21)
        .over("barrid")
        .alias("ff3_momentum")
    )
```

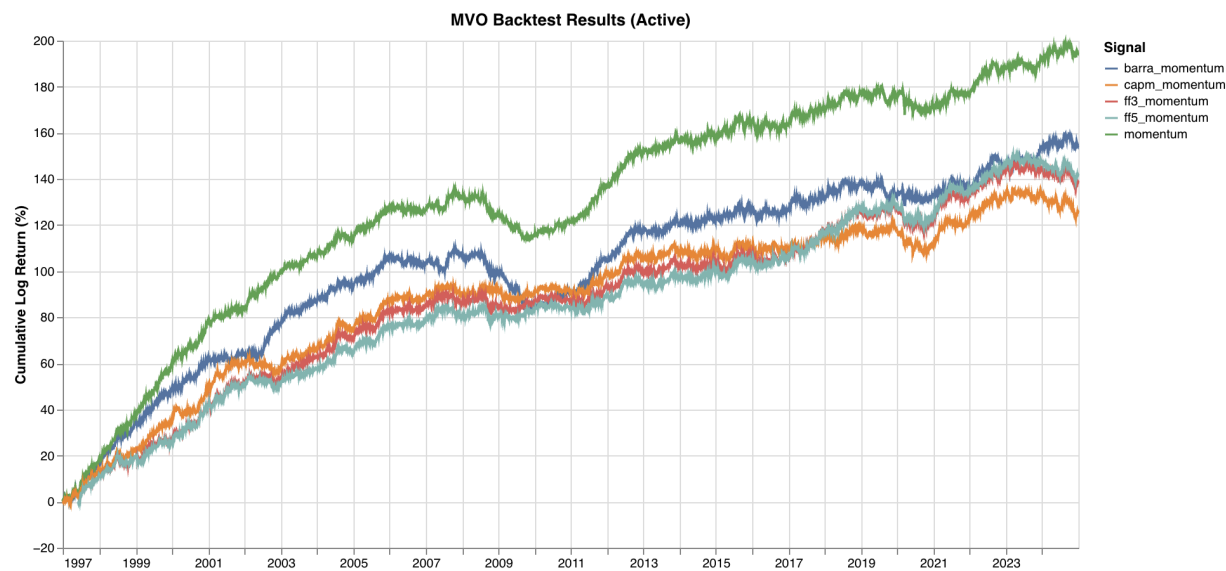
3. Barra Idiosyncratic Momentum Signal

```
def barra_idiosyncratic_momentum() -> pl.Expr:
    return(
        pl.col("specific_return")
        .log1p()
        .rolling_sum(230)
        .shift(21)
        .over("barrid")
        .alias('barra_momentum')
    )
```

4. Volatility-scaled Momentum

```
def volatility_scaled_momentum() -> pl.Expr:
    return (
        pl.col("return")
        .log1p()
        .rolling_sum(230)
        .truediv(pl.col("return").rolling_std(230))
        .shift(21)
        .over("barrid")
        .alias("vol_scaled_momentum")
    )
```

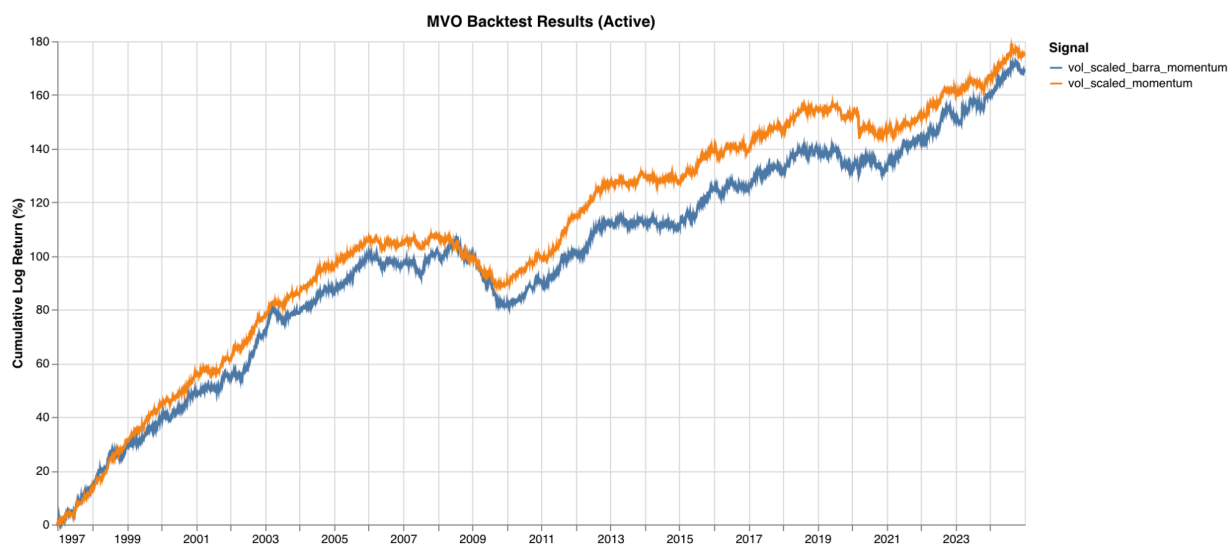
5. Full Sample Active Backtest Chart



6. Full Sample Active Backtest Table

MVO Backtest Results (Active)			
Signal	Mean Return	Volatility	Sharpe
barra_momentum	5.65%	5.08%	1.11
capm_momentum	4.61%	4.83%	0.95
ff3_momentum	5.15%	4.94%	1.04
ff5_momentum	5.26%	4.93%	1.07
momentum	7.10%	5.13%	1.38

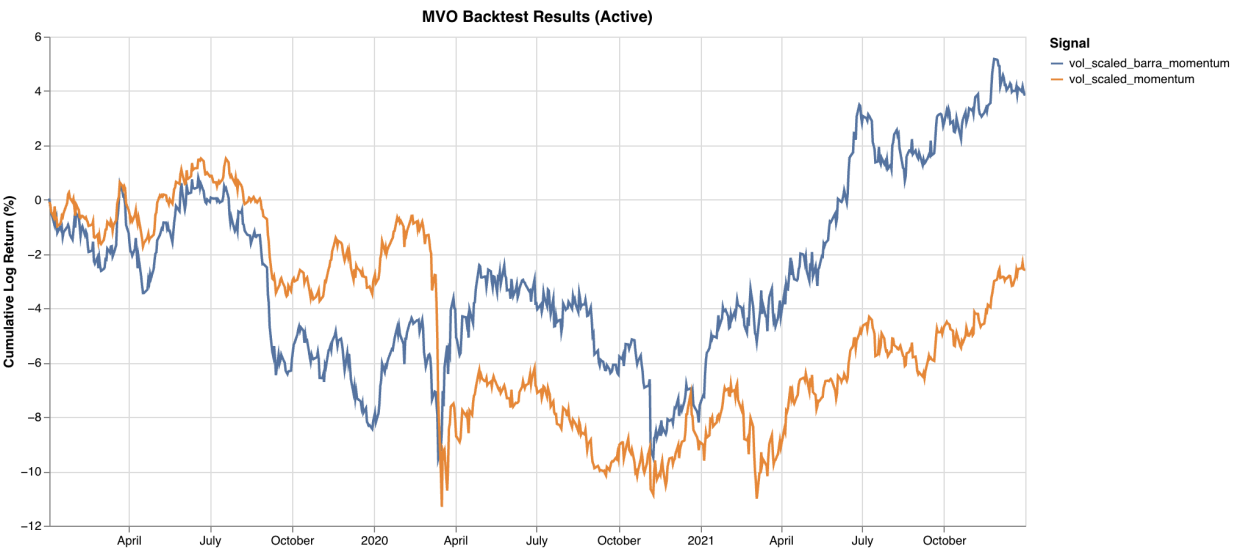
7. Full Sample Active Backtest Chart (Volatility-scaled)



8. Full Sample Active Backtest Table (Volatility-scaled)

MVO Backtest Results (Active)			
Signal	Mean Return	Volatility	Sharpe
vol_scaled_barra_momentum	6.17%	5.05%	1.22
vol_scaled_momentum	6.34%	4.01%	1.58

9. COVID-19 Sample Active Backtest Chart (Volatility-scaled)



10. COVID-19 Sample Active Backtest Table (Volatility-scaled)

MVO Backtest Results (Active)			
Signal	Mean Return	Volatility	Sharpe
vol_scaled_barra_momentum	1.47%	6.02%	0.24
vol_scaled_momentum	-0.71%	5.35%	-0.13